ADITYA UNIVERSITY

FUNDAMENTALS OF DATA SCIENCE (2501CS34)

(Common to CSE(DS))

UNIT-I				
1	Define Data Science.	L1	CO1	[2M]
2	Mention any two applications of Data Science.	L1	CO1	[2M]
3	Summarize the history and evolution of Data Science.	L2	CO1	[2M]
4	List the major stages of the Data Science Lifecycle.	L1	CO1	[2M]
5	Classify data into structured, semi-structured and unstructured data.	L3	CO1	[2M]
6	Mention any four tools and technologies used in Data Science.	L1	CO1	[2M]
UNIT-II				
1	Mention any two methods of data collection.	L1	CO2	[2M]
2	State the purpose of data cleaning.	L1	CO2	[2M]
3	Summarize any two methods for handling missing values.	L2	CO2	[2M]
4	Differentiate between normalization and standardization.	L2	CO2	[2M]
5	State the purpose of feature engineering.	L1	CO2	[2M]
6	Explain data integration and data aggregation.	L2	CO2	[2M]
UNIT-III				
1	Mention any two operations performed on text files.	L1	CO3	[2M]
2	State the purpose of reading text files in pieces.	L1	CO3	[2M]
3	Differentiate between JSON, XML and HTML data formats.	L2	CO3	[2M]
4	Summarize the concept of web scraping.	L2	CO3	[2M]
5	Mention the purpose of Binary and HDF5 file formats.	L1	CO3	[2M]
6	Identify the uses of Excel files, databases and Mongo DB in Data Science.	L1	CO3	[2M]
UNIT-IV				
1	Define Exploratory Data Analysis (EDA).	L1	CO4	[2M]
2	Mention any four descriptive statistical measures.	L1	CO4	[2M]
3	List any four data visualization techniques.	L1	CO4	[2M]
4	Differentiate between correlation and covariance.	L2	CO4	[2M]
5	Classify EDA into univariate, bivariate and multivariate analysis.	L3	CO4	[2M]
6	Summarize pattern detection and anomaly identification.	L2	CO4	[2M]
UNIT-V				
1	Mention the purpose of Neo4j.	L1	CO5	[2M]
2	State the basic components of a graph database.	L1	CO5	[2M]
3	Define Cypher.	L1	CO5	[2M]
4	Summarize the use of Cypher in Neo4j.	L2	CO5	[2M]
5	Mention the use of NLTK in text mining.	L1	CO5	[2M]
6	Differentiate between Neo4j and SQLite.	L2	CO5	[2M]